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EDUCATION

2018 - present

B. Eng. (in progress)
Tsinghua University

Expected June, 2022.

My major is an interdiscipline major. It lies in the intersection of Computer Science, Mechanical Engineering, and Electrical Engineering.

My minor is offered by School of Software, with **two graduate courses** attended:

- Information Visualization and Visual Analytics
- Deep Learning

2015 - 2018

High School Diploma
Xishan Senior High School

PROFILE

Affiliation

I am an **undergraduate** in **Tsinghua University**, China. I major in **Measurement, Control Technology and Instruments** (2018- present), with **Data Science and Technology Minor**.

Interests

My research interests lie primarily in robot control and machine Learning.

ACADEMIC & STANDARD TESTS

GPA and Rank

- Major: 3.89/4.00 (3/63, **top 5%**)
- Minor: 3.94/4.00 (no rank)

English Proficiency

- TOEFL: 106 (R30/L26/S22/W28)

GRE SCORE

- GRE: 328 (V158/Q170) + 4.0

PUBLICATION

My GOOGLE SCHOLAR

- **Zhang, C.**, Yu, W., & Li, Z. (2021). Accessibility-Based Clustering for Efficient Learning of Locomotion Skills. arXiv preprint arXiv:2109.11191. [preprint] [video]
  (Submitted to RAL+ICRA2022)
- **Zhang, C.**, Wu, Q., Ma, L., & Su, H. (2021). Adaptive Mimic: Deep Reinforcement Learning of Parameterized Bipedal Walking from Infeasible References. arXiv preprint arXiv:2112.03735. [preprint]
  (Submitted to L4DC2022)
- **Zhang, C.**, Hu, P., Zhang, R., Wu, Q., & Yang, A.. (2021). Modeling of Coupling Effects in Neural Networks for Ship Motion Prediction (Version 1). TechRxiv. https://doi.org/10.36227/techrxiv.14945544.v1 [preprint]

PROFESSIONAL ACTIVITIES

Review

Reviewer for 2022 American Control Conference.

RESEARCH EXPERIENCE

Robocup Humanoid Jun. 2021 - present

Developing the gait with my teammates

For the gait of our humanoid soccer TH-MOS, we are trying to implement DRL methods for high-speed and robust locomotion. Our group is co-advised by Dr. Li Liu and Dr. Jiwen Zhang.

Recently we work on imitatoin learning from low-quality references. We submitted our work to L4DC2022. For details, see: [preprint].

Efficient Learning of Robot Locomotion Skills Jun. 2021 - present

Summer intern, advised by Dr. Zhibin (Alex) Li.

Cooperating with Alex Li and Wanming Yu at Advanced Intelligent Robotics (AIR) Lab, University of Edinburgh, I have been engaged in studying efficient learning methods for robot locomotion skills, including fall recovery and backflipping.

We submitted our work to RAL+ICRA2022. For details, see: [preprint] [video]. We are also trying to extend our previous research now.

Modelling and Prediction of Ship Motions Oct. 2019 - Jun. 2021

Person in charge, advised by Dr. Peida Hu.

In our work, we managed to apply factorization machines to model the coupling effects of ship motions in the WD-RNN framework, thereby improving the prediction accuracy with low complexity. The proposed NN structure is also interpretable. For details, see: [preprint].

The program started when I entered my sophomore year, and the pandemic greatly encumbered the progress. Despite this, I managed to get top-level funding from the National Training Programs of Innovation and Entrepreneurship for Undergraduate (hosted by the Ministry of Education in China).

Topology Design for Multi-Chip Communication Jan. 2021 - Apr. 2021

Person in charge, advised by Dr. Guoqi Li.

This program was a follow-up of the previous research on the bio-inspired Tianjic chip. To achieve the number of neurons in human brain (86B), millions of chips (40k neurons per chip) are required, and the communication efficiency is quite a challenge. We investigated many existing topology structures for communication, and proposed a heuristic structure for high communication efficiency.

HONORS AND AWARDS

First Prize for Outstanding Student Research Training Project (2021). Best 10 among all of the undergraduate research projects at Tsinghua University.

Scholarship for Comprehensive Excellence Titled with December 9th. (2019). Highest level of scholarship in my major, 1 recipient out of 63 students.

Rank 6th in China Collegiate Computing Contest - Big Data Challenge 2020 (**national competition for college students**, with 2 teammates). The problem is to predict the ETA of the shipping orders.

SKILLS

Coding (mainly with Python/Matlab/C++), PCB design, embedded system development, mechanical design, Photoshop, office skills...

Also hobby: Pokémon Master

SOCIAL ACTIVITIES

Providing Equal Access to Engineering Training

I have been working in our department's **student association of technology** for three years, and currently I am the vice president. Our organization aims to develop students from diverse backgrounds who are interested in engineering to become experienced engineers. We provide funding, training and competitions for undergraduates, which are essential to improve their engineering skills. **Undergraduate students usually lack their own access to engineering training before entering labs for research, and it is hard for them to afford the cost, especially for those from poor families.**

I also developed my leadership and communication skills here.

COURSES

Math and Physics

Calculus, Linear Algebra, Probability and Statistics, Mathematical Modelling, Introduction to Complex Analysis, Physics, Theoretical Mechanics, Strength of Materials, Fundamentals of Engineering Thermodynamics.

Computer Science

Programming, Data Structure, The Fundamental of Computer: The Hardware/Software Interface, Introduction to Data Science, Information Retrieval Technology, Introduction to Big Data Systems, Fundamentals of Machine Learning, Intelligent Optimization Algorithms and Its Applications (attending), **Information Visualization and Visual Analytics** (graduate course), **Deep Learning** (graduate course).

Electrical Engineering

Introduction to Intelligent Connected System, Electrical Engineering and Electronics, Design and Practice of Circuit System, Special Experiments of Electromechanical System.

Mechanical Engineering

Fundamentals of Machine Design, Application of Mechanical Graphics, Creative Design in Mechanics, Manufacturing Engineering Fundamentals.

Instrumentation and Control

General Chemistry, Introduction of Modern Biology, Fundamentals of Optical Engineering, Error Theory and Data Processing, Fundamental Technology for Instruments Design, Measurement and Test Technology, Precision Instruments Design, Micro-electromechanical System Design, Measurement Circuits and Systems, Fundamentals of Control Engineering, Computer Control Technology, MEMS Technique (attending), Sensors and Signals (attending), Numerical Control Technology (attending).

*I prefer self-study and peer discussion rather than attending classes for new knowledge, so there are only a few graduate courses here. Btw, you can expect most of the courses listed here to be of grade 4.0.
*This list can be incomplete. It serves to make a representation of my inter-disciplinary background.